Exponentials & Polynomials

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Laws of Exponents & Fractional Powers

Simplify each expression using exponent and radical laws, assuming $x > 0$ and $y > 0$:

a $\frac{(2x^3y^{-2})^3}{4x^5y^{-4}}$

b $(27x^6)^{2/3} \cdot x^{-1}$

2 [Foundational] • [FRQ] • Target: Conjugates & Simplifying Radicals

Rationalize the denominator and simplify $\frac{12}{3-\sqrt{5}}$ into the form $a + b\sqrt{5}$.

3 [Foundational] • [FRQ] • Target: Polynomial Multiplication & Degree

Expand and simplify $(2x - 3)(x^2 + 4x - 5)$, and state the degree and leading coefficient of the resulting polynomial.

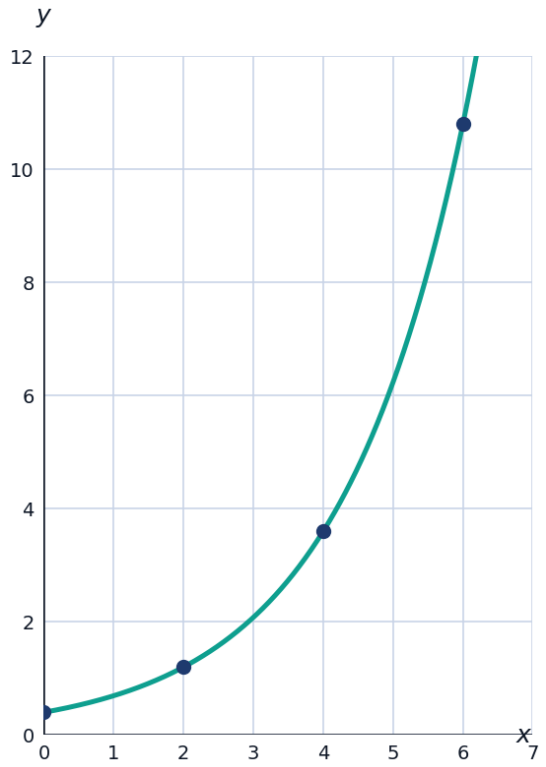
4 [Foundational] • [FRQ] • Target: Polynomial Remainder Theorem

Use the Remainder Theorem to find the remainder when $P(x) = 2x^3 - 5x^2 + 3x - 7$ is divided by $x - 2$.

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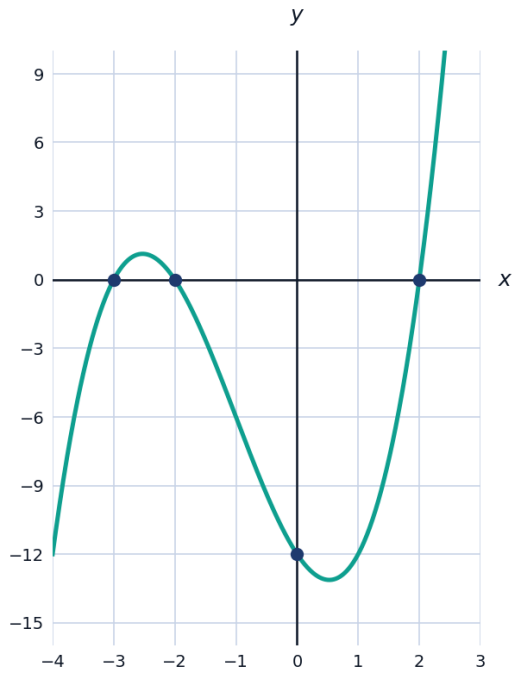
5 [Foundational] • [FRQ] • Target: Exponential Growth Modeling

A bacterial culture starts with 400 cells and triples every 2 hours. Write an exponential model $N(t)$ for the population after t hours, and calculate the population after 6 hours.



6 [Foundational] • [FRQ] • Target: Factoring Cubic Polynomials by Grouping

Factor the cubic polynomial $P(x) = x^3 + 3x^2 - 4x - 12$ completely by grouping and list all three real zeros.



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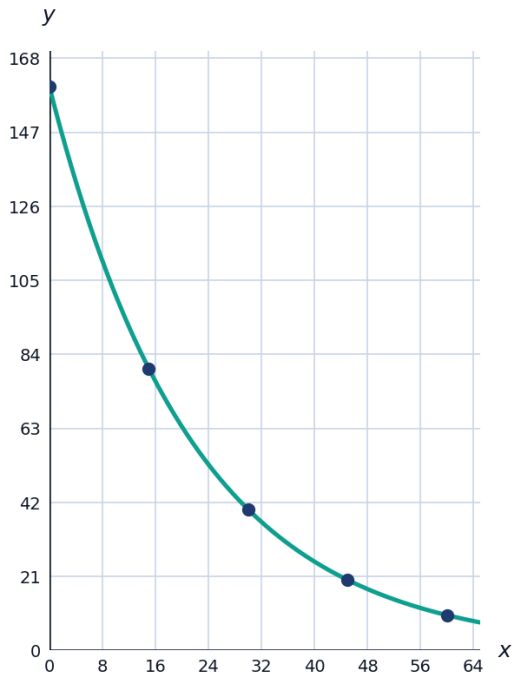
7 [Foundational] • [FRQ] • Target: Solving Exponential Equations with Common Bases

Solve the exponential equation $27^{2x-1} = 9^{x+2}$ by rewriting both sides with a common base.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Half-Life & Exponential Decay

A radioactive isotope has a half-life of 15 days. If a sample initially has a mass of $160\,\text{mg}$, write the decay formula $M(t)$ after t days and find how many days it takes for the sample to decay to $10\,\text{mg}$.



9 [Intermediate] • [FRQ] • Target: Radical Equations & Extraneous Solutions

Solve the radical equation $\sqrt{3x+7} - x = 1$ and verify which candidate root(s) are extraneous.

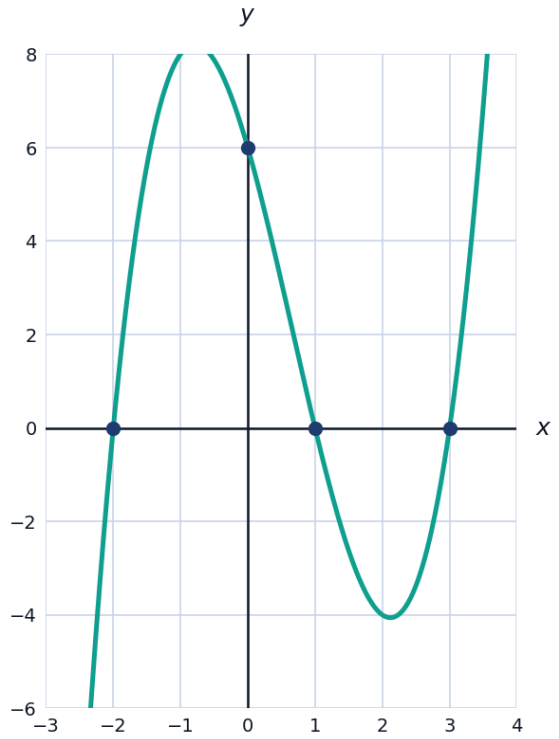
10 [Intermediate] • [FRQ] • Target: Factor Theorem & Polynomial Division

Given that $x + 2$ is a factor of $P(x) = 2x^3 + kx^2 - 7x - 6$, find the constant k and factor $P(x)$ completely.

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11 [Intermediate] • [FRQ] • Target: Polynomial Zeros, Multiplicity & Sign Analysis

The cubic polynomial $f(x) = (x + 2)(x - 1)(x - 3)$ is graphed below. State its x-intercepts, its y-intercept, and the intervals on which $f(x) > 0$.



12 [Intermediate] • [FRQ] • Target: Compound Interest & Effective Rate Conversion

An investment of $5{,}000$ dollars earns 8% annual interest compounded quarterly. Write the expression for its value $A(t)$ after t years, and find the effective annual percentage yield (APY) rounded to two decimal places.

13 [Intermediate] • [FRQ] • Target: Binomial & Cubic Algebraic Identities

If $x - \frac{1}{x} = 4$, compute the exact value of $x^2 + \frac{1}{x^2}$ and $x^3 - \frac{1}{x^3}$.

14 [Intermediate] • [FRQ] • Target: Rewriting Exponential Time Scales

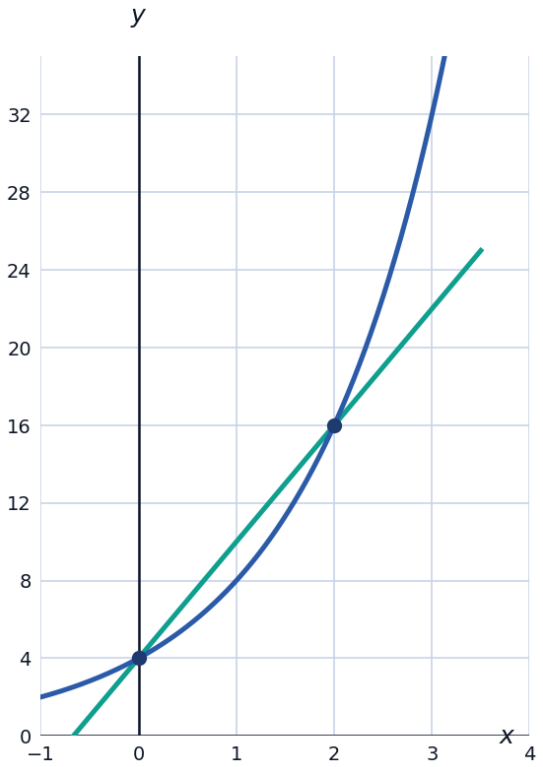
Rewrite the monthly growth model $V(t) = 1200(1.44)^{t/2}$ (where t is in years) in the form $V(t) = 1200(1 + r)^t$ and state the annual percentage growth rate r.

Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • [FRQ] • Target: Linear vs. Exponential Growth Comparison • Cross-Lesson: Topic 1.01 (Linear Models & Rate Comparison)

Compare the linear function $L(x) = 6x + 4$ and the exponential function $E(x) = 2^{x+2}$ at $x = 0, 1, 2, 3, 4$. Find all integer solutions to $L(x) = E(x)$ and state for which x we have $E(x) > L(x)$.



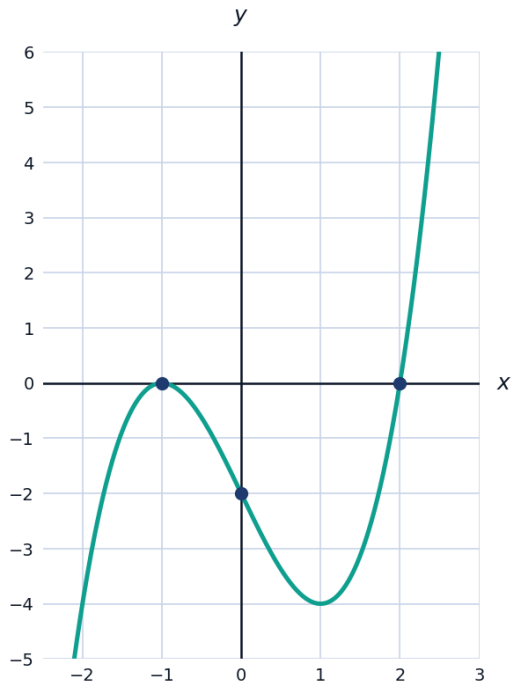
16 [Challenging] • [FRQ] • Target: Polynomial Division with Quadratic Divisors • Cross-Lesson: Topic 2.01 (Systems of Linear Equations)

A polynomial $P(x)$ leaves a remainder of 5 when divided by $x - 1$ and a remainder of -3 when divided by $x + 3$. Find the linear remainder $R(x) = ax + b$ when $P(x)$ is divided by $(x - 1)(x + 3)$.

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17 [Challenging] • [FRQ] • Target: Multiplicity & Tangency of Polynomial Graphs • Cross-Lesson: Topic 3.01 (Quadratic Tangency)

Let $f(x) = (x+1)^2(x-2)$. Find the coordinates of all x- and y-intercepts, identify at which intercept the graph is tangent to the x-axis vs. crosses it, and find $f(3)$.



18 [Challenging] • [FRQ] • Target: Systems of Exponential Equations • Cross-Lesson: Topic 2.01 (Linear Systems)

Solve the system of exponential equations $2^{x+y} = 32$ and $3^{2x-y} = 81$, and evaluate $x^2 + y^2$.

19 [Challenging] • [FRQ] • Target: Scaling Laws with Rational Exponents • Cross-Lesson: Topic 8.01 (3D Solids) & Topic 5.01 (Percentages)

A cube has surface area $S = 6s^2$ and volume $V = s^3$. Express the volume V as a function of surface area S using rational exponents, and find the percentage increase in V when S is multiplied by 4.

20 [Challenging] • [FRQ] • Target: Dual-Radical Equations • Cross-Lesson: Topic 3.01 (Quadratics)

Find all real solutions to $\sqrt{x+5} + \sqrt{x-3} = 4$.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Converting Radicals to Rational Exponents

For $x > 0$, which expression is equivalent to $\sqrt[3]{x^5} \cdot \sqrt{x^3}$?

a. $x^{8/5}$

b. $x^{5/2}$

c. $x^{15/6}$

d. $x^{19/6}$

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22 [Foundational] • [MCQ] • Target: Evaluating Exponential Expressions

If $3^{2x} = 81$, what is the value of 3^{x+1}?

a. 9

b. 18

c. 27

d. 84

23 [Foundational] • [MCQ] • Target: Polynomial Subtraction & Coefficient Sums

When $(3x^2 - 4x + 5) - (x^2 + 6x - 2)$ is written in the form $ax^2 + bx + c$, what is the value of $a + b + c$?

a. -1

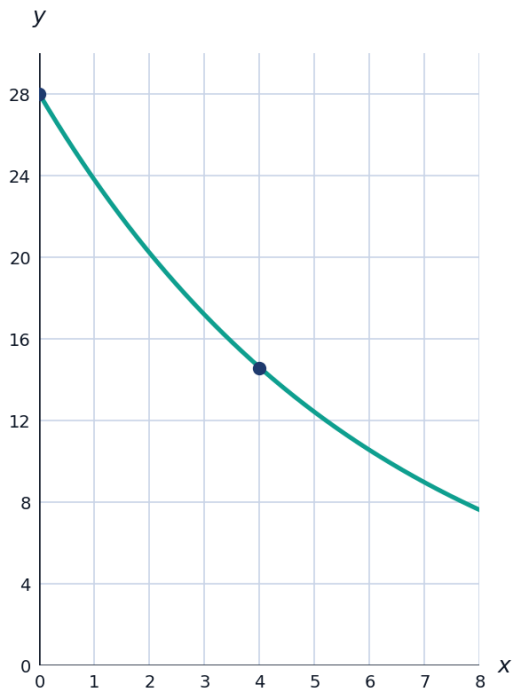
b. 3

c. 7

d. 11

24 [Foundational] • [MCQ] • Target: Exponential Decay Factor

A car purchased for $28{,}000$ dollars depreciates in value by 15% each year. Which function models the car's value $V(t)$, in dollars, t years after purchase?



a. $V(t) = 28000(0.15)^t$

b. $V(t) = 28000(0.85)^t$

c. $V(t) = 28000(1.15)^t$

d. $V(t) = 28000 - 0.15t$

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25 [Foundational] • [MCQ] • Target: Solving Basic Radical Equations

If $\sqrt{2x-3}=5$, what is the value of x?

a. 4

b. 11

c. 14

d. 28

26 [Foundational] • [MCQ] • Target: Difference of Cubes Factorization

Which of the following is a factor of x^3-27?

a. $x+3$

b. x^2-3x+9

c. x^2+3x+9

d. x^2-9

27 [Foundational] • [MCQ] • Target: Intercepts of Shifted Exponential Functions

What is the y-intercept of the exponential function $f(x)=5(2)^{x-1}-3$?

a. $(0,-0.5)$

b. $(0,2)$

c. $(0,5)$

d. $(0,7)$

28 [Foundational] • [MCQ] • Target: Identifying Exponential Tables

Which table of values represents an exponential function rather than a linear or quadratic function?

a. $(1,3),(2,6),(3,9),(4,12)$

b. $(1,3),(2,6),(3,12),(4,24)$

c. $(1,1),(2,4),(3,9),(4,16)$

d. $(1,5),(2,7),(3,11),(4,17)$

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

29 [Intermediate] • [MCQ] • Target: Exponent Base Unification

If $\frac{8^x}{2^y}=16$, what is the value of $3x-y$?

a. 2

b. 3

c. 4

d. 8

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30 [Intermediate] • [MCQ] • Target: Finding Coefficients with the Factor Theorem

When the polynomial $P(x) = x^3 - 4x^2 + kx + 6$ is divided by $x - 3$, the remainder is 0. What is the value of k?

a. -1

b. 1

c. 2

d. 3

31 [Intermediate] • [MCQ] • Target: Interpreting Fractional-Period Exponential Models

The population of a town is modeled by $P(t) = 18000(1.04)^{t/3}$, where t is the number of years after 2020. Which statement best interprets this model?

a. The population increases by 4% every year.

b. The population increases by 4% every 3 years.

c. The population increases by 3% every 4 years.

d. The population increases by 1.04% every 3 years.

32 [Intermediate] • [MCQ] • Target: Matching Polynomial Coefficients

If $(ax + 2)(bx - 3) = 12x^2 + cx - 6$ for all x, and $a + b = 7$, what are the two possible values of c?

a. -6 and -1

b. -4 and 1

c. -5 and 5

d. -2 and 2

33 [Intermediate] • [MCQ] • Target: Direct Fractional Power Scaling

For $x > 0$, if $x^{3/4} = 27$, what is the value of $x^{1/2}$?

a. 3

b. 9

c. 27

d. 81

34 [Intermediate] • [MCQ] • Target: Constructing Polynomials from Roots & Multiplicity

The polynomial $p(x)$ of degree 4 has a double root at $x = -2$, a simple root at $x = 1$, a simple root at $x = 3$, and $p(0) = 12$. What is $p(2)$?

a. -16

b. -8

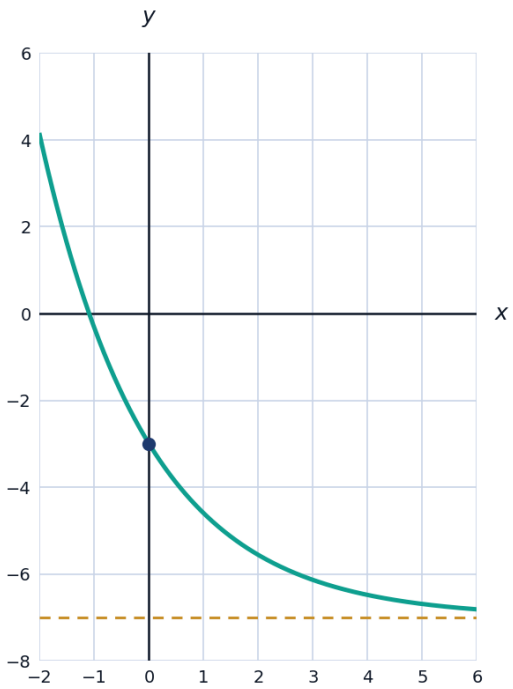
c. 8

d. 16

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35 **[Intermediate]** • **[MCQ]** • **Target: Asymptotes of Exponential Functions**

What is the horizontal asymptote of the graph of $f(x) = 4(0.6)^x - 7$ in the xy-plane?



a. $y = -7$

b. $y = -3$

c. $y = 0$

d. $y = 4$

36 **[Intermediate]** • **[MCQ]** • **Target: Combining & Re-Expressing Radicals**

If $\sqrt{50} + \sqrt{18} = \sqrt{k}$, what is the value of k?

a. 68

b. 96

c. 128

d. 144

37 **[Intermediate]** • **[MCQ]** • **Target: Sum of Cubes Identity**

If $x^3 + y^3 = 72$ and $x + y = 6$, what is the value of xy?

a. 6

b. 8

c. 10

d. 12

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Exponential Equations Reducible to Quadratics • Cross-Lesson: Topic 3.01 (Quadratics)

What is the sum of all real solutions to $9^x - 12 \cdot 3^x + 27 = 0$?

a. 3

b. 4

c. 12

d. 27

39 [Challenging] • [MCQ] • Target: Log-Linearized Exponential Models • Cross-Lesson: Topic 1.01 (Linear Equations) & Topic 6.01 (Data Modeling)

In a semi-log plot where $\log_{10}(y)$ is plotted on the vertical axis against x on the horizontal axis, the data points lie on the straight line $\log_{10}(y) = 2x + 3$. Which function gives y in terms of x?

a. $y = 1000(100)^x$

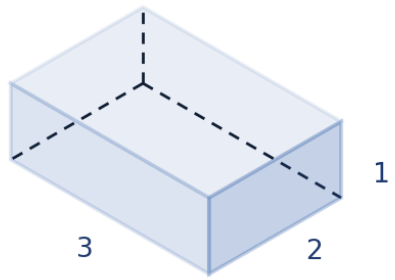
b. $y = 3(10)^{2x}$

c. $y = 100(1000)^x$

d. $y = 2x + 1000$

40 [Challenging] • [MCQ] • Target: Cubic Vieta's Formulas & 3D Surface Area • Cross-Lesson: Topic 8.01 (3D Solids & Surface Area)

The polynomial $P(x) = x^3 - 6x^2 + 11x - 6$ has three positive integer zeros that represent the lengths of the three edges of a rectangular prism. What is the surface area of the prism?



a. 6

b. 11

c. 22

d. 36

41 [Challenging] • [MCQ] • Target: Extraneous Root Filtering • Cross-Lesson: Topic 3.01 (Quadratics)

For how many real values of x does $\sqrt{x + 7} = x - 5$ hold?

a. Zero

b. One

c. Two

d. Three

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42 [Challenging] • [MCQ] • Target: Compound Percentage Scaling Across Multiples of Time • Cross-Lesson: Topic 5.01 (Percentages)

A savings account with continuous-style annual growth grows by 60% over every 4-year period. By what percentage does the account grow over an 8-year period?

a. 120%

b. 140%

c. 156%

d. 160%

43 [Challenging] • [MCQ] • Target: Polynomial Composition & Substitution • Cross-Lesson: Topic 10.01 (Function Composition)

If $P(x)$ is a polynomial such that $P(x^2+1) = x^4 + 6x^2 + 8$ for all real x, what is the value of $P(5)$?

a. 24

b. 35

c. 48

d. 63

44 [Challenging] • [MCQ] • Target: Coordinate Geometry of Polynomial Secants/Tangents • Cross-Lesson: Topic 1.01 (Linear Equations) & Topic 7.01 (Triangle Area)

What is the area of the triangle formed by the x-axis, the y-axis, and the tangent line to $y = x^2$ at $(0, 0)$ shifted to pass through $(2, 4)$ with slope 4?

a. 1

b. 2

c. 4

d. 8

45 [Challenging] • [MCQ] • Target: Comparing Large Exponentials via GCD Exponents • Cross-Lesson: Topic 2.01 (Inequalities & Ordering)

If $a = 2^{60}$, $b = 3^{45}$, and $c = 5^{30}$, which inequality correctly orders a, b, and c?

a. $a < c < b$

b. $a < b < c$

c. $c < a < b$

d. $b < c < a$

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